



Factoring Patterns

Polynomials are written in many different forms, some of which have recognizable factoring patterns.

Factoring the Greatest Common Monomial Factor

When all the terms of a polynomial have a common factor, you can use the distributive property to factor it out, so it is usually a good first thing to look for when factoring a polynomial.

EXAMPLE 1

Factor the greatest common monomial factor from $16x^3 - 12x^2 + 24x$.

SOLUTION

Step 1 Find the greatest common factor (GCF) of the polynomial. The GCF of the coefficients 16, 12, and 24 is 4, while the GCF of the variable expressions x^3 , x^2 , and x is x . So the GCF of the polynomial is $4x$.

Step 2 Factor out the GCF.

$$16x^3 - 12x^2 + 24x = 4x(4x^2 - 3x + 6) \quad \blacksquare$$

► **Think About It** Finding the GCF of a polynomial is really finding the GCF of the terms of the polynomial.

Factoring Special Patterns

You can use factoring patterns of special products to factor some polynomials.

Factoring Patterns

Pattern

Difference of Squares

$$a^2 - b^2 = (a - b)(a + b)$$

Perfect Square Trinomial

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Difference of Cubes

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Sum of Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Example

$$x^2 - 16 = (x - 4)(x + 4)$$

$$x^2 + 10x + 25 = (x + 5)^2$$

$$x^2 - 10x + 25 = (x - 5)^2$$

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$$

$$x^3 + 8 = (x + 2)(x^2 - 2x + 4)$$

EXAMPLE 2

Factor.

A $9x^2 - 121$

SOLUTION

The binomial $9x^2 - 121$ can be written as $(3x)^2 - 11^2$ and can be factored with the difference of squares pattern.

$$9x^2 - 121 = (3x - 11)(3x + 11)$$

► **Think About It** The difference of two squares is a binomial that factors into the product of two binomials.

B $x^3 - 64$

SOLUTION

The binomial $x^3 - 64$ can be written as $x^3 - 4^3$ and can be factored with the difference of cubes pattern.

$$x^3 - 64 = (x - 4)(x^2 + 4x + 16)$$

C $x^2 - 14x + 49$

SOLUTION

The trinomial $x^2 - 14x + 49$ can be written as $x^2 - 2 \cdot 7x + 7^2$ and can be factored with the perfect square pattern.

$$x^2 - 14x + 49 = (x - 7)^2$$

D $2x^3 + 64x^2$

SOLUTION

The binomial $2x^3 + 64x^2$ has a common monomial $2x^2$ that can be factored out.

$$2x^3 + 64x^2 = 2x^2(x + 32)$$

E $x^2 + 36$

SOLUTION

The binomial $x^2 + 36$ is a sum of two squares that cannot be factored.

► **Think About It** The sum of two squares cannot be factored with real numbers.

F $x^3 + 125$

SOLUTION

The binomial $x^3 + 125$ can be written as $x^3 + 5^3$ and can be factored with the sum of cubes pattern.

$$x^3 + 125 = (x + 5)(x^2 - 5x + 25) \quad \blacksquare$$

Factoring Trinomials of the Form $x^2 + bx + c$

When you use the FOIL method to multiply two binomials $(x + s)(x + t)$, you can write the product as follows:

$$x^2 + tx + sx + st = x^2 + (t + s)x + st$$

Some trinomials of the form $x^2 + bx + c$ can be factored as

$$\begin{aligned}x^2 + \textcolor{blue}{b}x + \textcolor{blue}{c} &= x^2 + (\textcolor{red}{t} + \textcolor{blue}{s})x + \textcolor{blue}{st} \\ &= (x + s)(x + t)\end{aligned}$$

where $\textcolor{red}{t} + \textcolor{blue}{s} = \textcolor{blue}{b}$ and $\textcolor{blue}{st} = \textcolor{blue}{c}$.

You can use this pattern to help you with a guess-and-check strategy to factor trinomials of the form $x^2 + bx + c$.

► **Think About It** If there are no values for s and t with product c and sum b , then the polynomial can't be factored. Any polynomial that can't be factored is a **prime polynomial**.

EXAMPLE 3

Factor.

A $x^2 + 15x + 36$

SOLUTION

In this trinomial, b and c are both positive, so s and t are both positive. Find the solution by “guessing” the factors of 36 and checking to determine which factor pair has a sum of 15.

$$x^2 + 15x + 36 = (x + 3)(x + 12)$$

Positive factors of 36	Sum of the factors
1, 36	37
2, 18	20
3, 12	15
4, 9	13
6, 6	12

B $x^2 + 5x - 14$

SOLUTION

In this trinomial, c is negative, so s and t have opposite signs. Find the solution by “guessing” the factors of -14 and checking to determine which factor pair has a sum of 5.

$$x^2 + 5x - 14 = (x - 2)(x + 7)$$

Opposite-sign factors of -14	Sum of the factors
1, -14	-13
-1 , 14	13
-2, 7	5
2, -7	-5

C $x^2 - 6x - 7$

SOLUTION

In this trinomial, c is negative, so s and t have opposite signs. Find the solution by “guessing” the factors of -7 and checking to determine which factor pair has a sum of -6 .

$$x^2 - 6x - 7 = (x + 1)(x - 7)$$

Opposite-sign factors of -7	Sum of the factors
1, -7	-6
-1 , 7	6

D $x^2 - 13x + 22$

SOLUTION

In this trinomial, b is negative and c is positive, so s and t are both negative. Find the solution by guessing the factors of 22 and checking to determine which factor pair has a sum of -13 .

$$x^2 - 13x + 22 = (x - 2)(x - 11)$$

Negative factors of 22	Sum of the factors
-1 , -22	-23
-2, -11	-13

E $x^2 - 7x + 9$

SOLUTION

In this trinomial, b is negative and c is positive, so s and t are both negative. Write the negative factors of 9 and check to determine which pair has a sum of -7 .

None of the factor pairs have a sum of -7 , so this trinomial cannot be factored and is prime.

Negative factors of 9	Sum of the factors
-1 , -9	-10
-3 , -3	-6